

Peter Pao-Huang

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EDUCATION

Stanford University

Ph.D. in Computer Science, advised by Stefano Ermon and Xiaojie Qiu

Palo Alto, California

Sep 2024 – May 2029

University of Illinois at Urbana-Champaign

B.S. in Computer Science

Champaign, Illinois

Sep 2020 – May 2024

PUBLICATIONS

Pao-Huang, P., Qiu, X., Ermon, S., 2026. Generative Modeling with Flux Matching. *In Conference Submission*.

Pao-Huang, P., Black, M., Qiu, X., 2025. Geometric Generative Modeling with Noise-Conditioned Graph Networks. *International Conference on Machine Learning*.

Pao-Huang, P., Thrush, K., Montemayor, D., Levine, M., 2023. Unpaired Single Cell Dataset Alignment with Wavelet Optimal Transport. *In Conference Submission*.

Pao-Huang, P., Jing, B., Berger, B., 2023. Scalable Multimer Structure Prediction using Diffusion Models. *NeurIPS 2023 AI4Science Workshop*.

Jing, B., Erives, E., **Pao-Huang, P.**, Corso, G., Berger, B. and Jaakkola, T., 2023. EigenFold: Generative Protein Structure Prediction with Diffusion Models. *ICLR MLDD Workshop*.

Zhao, Y., Sharif, H., **Pao-Huang, P.**, Shah, V., Sivakumar, A. N., Valverde Gasparino, M., ... & Adve, V. (2023). ApproxCaliper: A Programmable Framework for Application-aware Neural Network Optimization. *Proceedings of Machine Learning and Systems*.

RESEARCH EXPERIENCE

Altos Labs

Research Intern, Advised by Professor Morgan Levine

San Diego, California

Jun 2023 – Oct 2023

- Worked with the Morgan Levine group on generative modeling efforts for single-cell omics and imaging in cellular rejuvenation.
- Created a novel optimal transport method (“Wavelet Optimal Transport”) for unpaired modality alignment using spectral graph wavelets.

Computation & Biology Group (MIT CSAIL)

Research Assistant, Advised by Professor Bonnie Berger

Cambridge, Massachusetts

Oct 2022 – Jun 2023

- Researched generative modeling and physics-informed deep learning applied to drug discovery.
- Developed a novel diffusion-based generative model for protein conformation sampling and folding.
- Created a new diffusion SE(3)-equivariant model to predict large multimeric structures that is more scalable than previous methods.

LLVM Research Group

Research Assistant, Advised by Professor Vikram Adve

Urbana, Champaign

Sept 2020 – Sept 2022

- Co-authored paper published in MLSys for neural network optimization (structured pruning, low-rank factorization, and quantization) and approximate computing on edge computing devices.
- Developed PyTorch-based federated semi-supervised learning framework for distributed, computer vision-based systems.

WORK EXPERIENCE

Two Sigma Investments

Software Engineering Intern

New York, New York

Jun 2022 – Aug 2022

- Integrated Bayesian statistical modeling to critical classes of forecast models.
- Built framework for evaluating probabilistic programming languages (PPLs) like PyMC and Stan, used by quantitative researchers to select the optimal language to express Bayesian models.

Avant

Machine Learning Intern

Chicago, Illinois

May 2021 – Aug 2021

- Developed a Python-based machine learning library utilizing distributed computing to enable model training and scoring on large-scale datasets.
- Deployed library through Docker on production Kubernetes clusters used for in-house model training and evaluation tasks.
- Created parallelized versions for Scikit-learn models, transforms, & hyperparameter optimization

AWARDS & FELLOWSHIPS

Stanford Bio-X Fellowship	2026–2029
National Science Foundation Graduate Research Fellowship (NSF GRFP)	2025–2028
Stanford School of Engineering Fellowship	2024–2025